

Tariffs and the Secondhand Shift

Price Pass-Through and Substitution Between New and Used Goods

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1 Motivation and Research Question

1.1 Tariff Reform and Core Research Question

In 2025, the United States implemented a global tariff package, significantly affecting the consumer price of many goods and increasing friction in trade between different countries. In particular, we can see how the following were imposed: (i) Basic duties applied extensively to most products, (ii) Elimination of the de minimis exemption for low-value parcels, (iii) Rules that increasingly require Delivery Duty Paid for postal shipments. The above rules raise the question: will consumers switch from new goods to used/refurbished goods when the prices of new goods increase? And, if so, by how much?

1.2 Research Objective and Hypotheses

Research objective. After the 2025 tariff changes, how much of this shock will be absorbed by sellers of used/refurbished goods compared to sellers of new goods in major categories such as clothing, electronics and cars? Will consumers make more use of second-hand goods?

Testable hypotheses

1. **H1 (Pass-through asymmetry).** Post-policy price increases are greater for **new goods** than for **used/refurbished** goods of the same quality.
2. **H2 (Substitution).** The **share of spending** on used goods increases after the dates of the duties, particularly in **high durability** categories (cars, some electronics).

1.3 Economic mechanisms and opposing forces

Economic mechanisms and opposing forces. The supply and demand model predicts that a rise in the prices of new goods will lead to their replacement with used/refurbished goods when these are comparable in quality. Three factors complicate the picture:

- **Supply frictions in the second-hand market:** If households delay buying new products due to rising prices, they will have fewer goods to put back on the second-hand market in the short term.
- **Cross-border frictions:** DDP and tariffs on small parcels make buying second-hand goods from abroad more expensive and slower.

- **Substitution effect:** How well second-hand goods replace new ones depends on how long the goods last and how their quality changes. For example, seasonal gadgets have low substitutability, while cars and laptops have high substitutability. Therefore, we expect a greater shift where products hold up well.

1.4 Related Literature and Contribution

Previous Literature: Public debate around the 2025 tariff package suggests higher trade barriers would increase the price of new goods and could push consumers towards resale and refurbished channels. Industry reports document steady growth for second-hand platforms in the run-up to 2025 and during recessions, when thinner budgets increase the attractiveness of used goods. Academic work has studied tariff pass-through into import and consumer prices, and a separate literature has examined second-hand markets using matching and hedonic models. Still, little causal evidence exists on how trade shocks propagate into second-hand markets, either via substitution between new and used goods or via differential pass-through into new versus used prices; most empirical studies of tariffs focus only on prices of new goods.

Our Contribution: There are three primary ways that this paper contributes. First, it quantifies the degree to which used and refurbished goods absorb tariff-driven price increases and offers causal evidence on how a significant trade policy shock is distributed between new and second-hand markets. Second, it records how buyers and sellers modify a variety of margins, such as prices, listing volumes, used shares, and second-hand search and disposal behaviour, and how these modifications vary among product categories with different durability. Third, by comparing the United States to a set of donor countries, it offers a cross-country perspective on the role of second-hand markets as a buffer against trade shocks and highlights the value of incorporating second-hand price and activity measures alongside official statistics when assessing the incidence of trade policy.

2 Data

2.1 Countries, Time Period, and Product Categories

The analysis focuses on the United States. In addition, a small group of economies, such as Canada and the United Kingdom, are used as a set of control or donor countries. The treatment is conducted in the period after Trump’s tariff announcement on 2 April 2025. In fact we are going to consider April as the implementation month in Methodology.

We focus on three product categories, indexed by c , differentiated by substitution effect:

- Fast-fashion clothing ($c = A$)
- Portable consumer electronics such as smartphones and laptops ($c = E$)
- Passenger cars ($c = K$)

2.2 Panel Structure and Notation

After defining the time horizon, countries and research products, we define the two main panels that will be used in the empirical analysis and how they link directly with H1 and H2:

(i) Prices Panel (Defined for each U.S. region $\times$ category $\times$ month panel): Within the United States, listings are aggregated to broad regions $g \in \{\text{Northeast, Midwest, South, West}\}$ and months m . For each (c, g, m) we define:

- $\log \hat{P}_{c,g,m}^{\text{used}}$: logarithm of quality-adjusted used/refurbished price index, obtained from the hedonic ML model applied to the listing data described in Section 2.2.
- $\log P_{c,m}^{\text{new}}$: logarithm of official new-goods price index (BLS / national statistical office CPI category indices for new goods.) for category c in month m (and, for cars, official used-vehicle indices where available).
- $L_{c,g,m}^{\text{used}}$: number of used listings in category c posted in region g during month m (a simple count of second-hand items on the platforms).
- $S_{c,g,m}^{\text{used}}$: a simple used-share proxy, e.g. based on $L_{c,g,m}^{\text{used}}$ relative to total listings or to a pre-policy baseline. For example,

$$S_{c,g,m}^{\text{used}} = \frac{L_{c,g,m}^{\text{used}}}{L_{c,g,m}^{\text{total}}},$$

where $L_{c,g,m}^{\text{total}}$ is the total number of listings (new + used).

The Price Panel is used in H1 in order to compare changes in $\log \hat{P}_{c,g,m}^{\text{used}}$ and $\log P_{c,m}^{\text{new}}$ around m^* , and in H2 to study post-policy changes in $S_{c,g,m}^{\text{used}}$.

(ii) Behavioral panel (Defined for each Country $\times$ category $\times$ week panel): Countries are indexed by r (United States and donors), categories by $c \in \{A, E, K\}$, and weeks by t . For each (r, c, t) :

- $Y_{r,c,t}^{\text{SH}}$: Secondhand Demand Index, built from standardized Google Trends queries reflecting *demand* for used/refurbished goods in category c (e.g., “used iPhone”, “thrift store”, “used car”, plus platform names).
- $Y_{r,c,t}^{\text{SS}}$: Secondhand Supply Index, built from queries reflecting *supply* or disposal (e.g., “sell my phone”, “sell clothes online”, “donate clothes”).

The Behavioral Panel is used in H2 to compare U.S. secondhand demand/supply to donors around t^* .

Note that the Price Panel is defined by month, as most data on official price changes are released on a monthly basis. Furthermore, official new goods are not defined for each region, as official data are defined at a national level, while we prefer to have used products defined for each region, as there is more variability.

Notation summary. For convenience, the main indices are:

Symbol	Meaning
c	Product category ($c \in \{A, E, K\}$)
g	U.S. region ($g \in \{\text{Northeast, Midwest, South, West}\}$)
r	Country (United States or donor country)
m	Calendar month (Price Panel)
t	Calendar week (Behavioral Panel)

3 Methodology

3.1 Overview of Empirical Strategy

After constructing the panel datasets described in Section 2, we outline how they are used to test H1 and H2. The empirical strategy proceeds in two main steps:

1. **H1 (Pass-through asymmetry).** Using the U.S. region $\times$ category $\times$ month Price Panel, we estimate event-study regressions for $\log P_{c,m}^{\text{new}}$ and $\log \hat{P}_{c,g,m}^{\text{used}}$ around the 2025 tariff dates. Comparing the post-policy paths of these coefficients across categories provides evidence on whether quality-adjusted prices of new goods rise more than those of used or refurbished goods.
2. **H2 (Substitution into used goods).** We study substitution along two dimensions:
 - *Listing-based substitution (within the U.S.).* Event-study regressions for $S_{c,g,m}^{\text{used}}$ and $L_{c,g,m}^{\text{used}}$ measure changes in the relative importance of used listings across regions and categories after the tariffs.

- *Behavioral substitution (U.S. vs. donors).* Using the country $\times$ category $\times$ week Behavioral Panel, we estimate Diff-in-Diff event studies for $Y_{r,c,t}^{\text{SH}}$ and $Y_{r,c,t}^{\text{SS}}$ to compare U.S. secondhand demand and supply to donor countries around the policy dates.

3.2 Construction of Key Indices

Goal. Summarize how the main indices from Section 2 are constructed before they enter the causal analysis.

Hedonic ML indices for used prices ($\log \hat{P}_{c,g,m}^{\text{used}}$)

We construct quality-adjusted used-price indices from a hedonic model. For each used item i with price P_i , characteristics X_i , region g , category $c \in \{A, E, K\}$, and month m , we estimate

$$\log P_i = \alpha_{c,g,m} + f_c(X_i) + \varepsilon_i,$$

where $\alpha_{c,g,m}$ are region $\times$ category $\times$ month effects and $f_c(\cdot)$ is a flexible prediction function estimated separately by category using gradient boosting regression. We use gradient-boosting regression because it handles nonlinearities and interactions between features, such as model $\times$ mileage, and usually performs very well on tabular data. The feature vector X_i includes standard product attributes, e.g. for cars: make, model, model year, mileage, fuel type, transmission, trim) and a harmonized condition measure obtained by mapping platform-specific labels and free-text descriptions into a common ordered scale (new with tags/like new/good/fair/for parts) with the with the help of a language model, validated on a small hand-labeled subset.

To limit overfitting, $f_c(\cdot)$ is estimated with K -fold cross-fitting. Using the resulting out-of-fold predictions $\hat{f}_c(X_i)$, we first construct residuals

$$\tilde{y}_i = \log P_i - \hat{f}_c(X_i),$$

and then estimate

$$\tilde{y}_i = \alpha_{c,g,m} + u_i,$$

where $\alpha_{c,g,m}$ are region $\times$ category $\times$ month dummies (and no additional intercept). Let $\hat{\alpha}_{c,g,m}$ denote the estimated fixed effects. For each (c, g) , we normalize these to a pre-policy base month m_0 by defining

$$\log \hat{P}_{c,g,m}^{\text{used}} = \hat{\alpha}_{c,g,m} - \hat{\alpha}_{c,g,m_0},$$

and report the level index as

$$\hat{P}_{c,g,m}^{\text{used}} = 100 \times \exp(\log \hat{P}_{c,g,m}^{\text{used}}),$$

so that $\hat{P}_{c,g,m_0}^{\text{used}} = 100$ by construction. The series $\log \hat{P}_{c,g,m}^{\text{used}}$ is the quality-adjusted used price index entering the event-study regressions for H1 and H2.

Used volume ($L_{c,g,m}^{\text{used}}$) and used share ($S_{c,g,m}^{\text{used}}$)

For each category c , region g , and month m , we construct two simple measures from the platform data. First, $L_{c,g,m}^{\text{used}}$ is the number of secondhand items in category c that are offered for sale in region g during month m . Second, we define a used-share measure

$$S_{c,g,m}^{\text{used}} = \frac{L_{c,g,m}^{\text{used}}}{L_{c,g,m}^{\text{total}}},$$

where $L_{c,g,m}^{\text{total}}$ is the total number of items (new and secondhand) observed in that cell. Both $L_{c,g,m}^{\text{used}}$ and $S_{c,g,m}^{\text{used}}$ are later used as outcomes in the substitution analysis for H2.

Behavioral demand ($Y_{r,c,t}^{\text{SH}}$) and supply indices ($Y_{r,c,t}^{\text{SS}}$)

For each country r and category c , we construct indices from Google Trends as follows:

1. select a set of queries related to secondhand *demand* (e.g. “used iPhone”, “thrift store”, “used car”, plus platform names) and *supply* (e.g. “sell my phone”, “sell clothes online”, “donate clothes”),
2. standardize each query series over time,
3. apply principal components analysis (PCA) and take the first principal component as the Secondhand Demand Index $Y_{r,c,t}^{\text{SH}}$ or the Sell-Side Index $Y_{r,c,t}^{\text{SS}}$.

3.3 H1: Pass-Through Asymmetry Between New and Used Goods

Goal. Measure how strongly the prices of new and used goods react to the 2025 tariff package, and compare them. For H1, we work with the U.S. region $\times$ category $\times$ month Price Panel. Let i index a region–category cell and t index months. Define:

$$Y_{i,t}^U = \log \hat{P}_{i,t}^{\text{used}}, \quad Y_{i,t}^N = \log P_{i,t}^{\text{new}},$$

and let $D_{k,t}$ be a dummy equal to one if month t is k months away from the implementation month t^* (i.e. $t - t^* = k$), with $k = -1$ omitted as the reference period. The event-study

regressions are:

$$Y_{i,t}^U = \sum_{k \neq -1} \beta_k^U D_{k,t} + \mu_i + \tau_t + u_{i,t},$$

$$Y_{i,t}^N = \sum_{k \neq -1} \beta_k^N D_{k,t} + \mu_i + \tau_t + v_{i,t},$$

where μ_i are region–category fixed effects and τ_t are month fixed effects.

Interpretation (H1). The coefficients β_k^U and β_k^N trace the dynamic response of used and new prices around the tariff dates, relative to month $k = -1$. H1 is supported if, for tariff-exposed categories, the post-policy coefficients β_k^N are systematically larger than β_k^U , i.e. if quality-adjusted prices of new goods rise more than those of used or refurbished goods.

3.4 H2: Substitution Toward Used Goods

Goal. Detect whether consumers shift towards used goods after the tariffs, using both platform-based measures and behavioral indices.

(a) Listing-based substitution (U.S. regions). Using the same region–category–month panel, we study the used-share measure $S_{i,t}^{\text{used}}$ and the used volume $L_{i,t}^{\text{used}}$. With the same event-time dummies $D_{k,t}$ as in H1, a simple event-study specification for the used share is:

$$S_{i,t}^{\text{used}} = \sum_{k \neq -1} \phi_k D_{k,t} + \mu_i + \tau_t + e_{i,t},$$

with region–category fixed effects μ_i and month fixed effects τ_t . Positive post-policy coefficients ϕ_k , especially for more durable categories, indicate an increase in the relative importance of used goods.

Interpretation for: The used-share response supports H2 if the coefficients ϕ_k turn positive. and remain high in the post-policy months thereby suggesting that U.S. regions list or a greater share of second-hand goods after the tariffs. Larger effects for durable categories would be consistent with consumers substituting away from tariff-affected new goods towards the used market.

(b) Behavioral substitution (U.S. vs. donors). For the behavioral panel (country $\times$ category $\times$ week), we use weekly indices of secondhand demand and supply, $Y_{r,c,t}^{\text{SH}}$ and $Y_{r,c,t}^{\text{SS}}$. Let US_r be a dummy equal to one for the United States and zero for donor countries, and let $D_{k,t}^{\text{week}}$ be event time dummies defined relative to the policy week t^* . A compact event-study

DiD specification for demand is:

$$Y_{r,c,t}^{\text{SH}} = \sum_{k \neq -1} \beta_k (D_{k,t}^{\text{week}} \cdot US_r) + \gamma_{r,c} + \delta_t + \varepsilon_{r,c,t},$$

where $\gamma_{r,c}$ are country–category fixed effects and δ_t are week fixed effects. Analogous regressions are run for $Y_{r,c,t}^{\text{SS}}$.

Interpretation for: The behavioral indices support H2 if the post-policy coefficients β_k are positive, reflecting that U.S. secondhand demand and supply increase more strongly than in the donor countries following tariffs. A systematic divergence in the U.S. relative to donors, especially in durable categories, would reflect a behavioural shift towards used goods in response to the higher prices of new goods.

3.5 Robustness, Placebo Tests, and Threats to Identification

Robustness. By varying the duration of the event window, the set of control countries, the set of control variables, and the sample definition (including alternative trimming rules), we will assess the robustness of our findings. The objective is to confirm that our findings are independent of a particular modelling decision or data subset.

Falsification. Using alternative donor or synthetic-control constructions, placebo dates, and placebo categories, we will conduct a number of falsification tests. Placebo categories are product groups that have little exposure to the 2025 tariff package, and placebo dates move the policy intervention to times when there are no tariff changes. Significant "effects" in these placebo settings or the lack of differences in the primary specifications between the US and control nations would be seen as proof against a tariff-driven transition from new to used goods.

What could distort the analysis. The estimated effects could still be distorted by a number of factors. First, the 2025 tariff package may be mistakenly blamed for concurrent policy changes or macroeconomic shocks that have a different impact on new versus used goods but are not fully captured by our controls. Second, measurement error in platform data (e.g., changes in search algorithms, reporting procedures, or platform composition over time) may attenuate true effects or introduce spurious breaks, particularly if these changes vary between countries or product categories. Third, anticipatory actions by buyers and sellers, like stockpiling or making calculated price changes prior to the official implementation date, may cause responses to be delayed beyond the predetermined event window.

Reproducibility. The primary results and specifications will be pre-registered, code and appropriately anonymised or synthetic data will be released, and cleaning rules and index construction documentation will be documented.

Ethics and data use. Only public pages or APIs are used to gather data, and robots.txt and platform rate limits are respected. Geographical information is kept at an aggregate regional level, no personal identifiers are stored, and I rely on official price indices and behavioural proxies for categories where platform data is not trustworthy.

4 Expected Results

4.1 Expected Quantitative Patterns

Goal. Summarize, in a simple way, the main quantitative patterns that the empirical strategy is designed to detect.

- **H1 (Pass-through).** After the 2025 tariffs, the quality-adjusted price index for new goods is expected to rise systematically more than the corresponding index for used or refurbished goods. In practice, this means that the post-policy coefficients for new prices in the event-study plots should lie clearly above those for used prices in tariff-exposed categories, indicating stronger pass-through into new-goods prices.
- **H2 (Substitution).** Following the tariffs, the relative importance of used goods is expected to increase. Quantitatively, the used share $S_{i,t}^{\text{used}}$ and used volume $L_{i,t}^{\text{used}}$ should rise in the U.S. region–category–month panel, while the behavioral indices $Y_{r,c,t}^{\text{SH}}$ and $Y_{r,c,t}^{\text{SS}}$ should show a sustained upward deviation for the U.S. relative to donor countries, especially in durable categories such as cars and some electronics.

4.2 Implications and Takeaways for Readers

Taken together, the results should show: i) how much of the tariff shock is absorbed by new versus used prices and whether, and in what categories, consumers turn to second-hand or refurbished goods. If both hypotheses are supported, the analysis argues that enhancing resale infrastructure can partly cushion consumers from trade shocks, and that quality-adjusted indices for second-hand markets could play a useful role along with official price statistics and satellite accounts.

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